

3. Simulating a system of N particles

- How to analyze the results:
 - ✓ Structure: the radial distribution function
 - ✓ Dynamic properties: the velocity autocorrelation function

3. Simulating a system of N particles

- Structure: The radial distribution function (RDF)

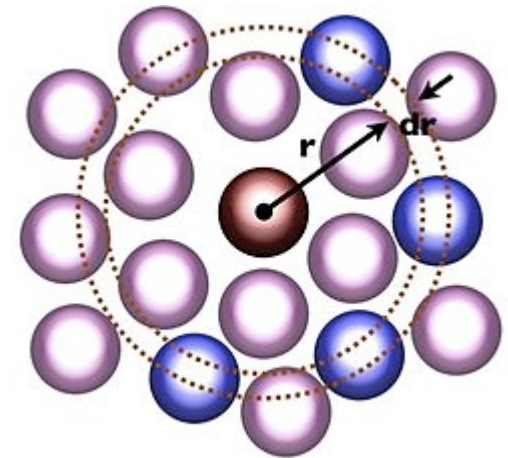
- The radial distribution function (RDF) describes how density varies as a function of distance from a reference particle. It can be defined as

$$g(r) = \frac{1}{\rho} \frac{n(r)}{4\pi r^2 dr} = \frac{n(r)}{n_{id}(r)}$$

$\rho = \frac{N}{V}$ average number density of particles

$n(r)$ Mean number of particles within a distance of r and $r+dr$ away from a particle.

$n_{id}(r)$ Mean number of particles for an ideal gas



calculation of $g(r)$

- The RDF is usually determined by calculating the distance between all particle pairs and binning them into a histogram.

Algorithm for the calculation of the RDF

Initialization, goes before main loop of the MD code

```
delg=box/(2*nhis)
do i = 1,nhis
  g(i) = 0
enddo
```

bin size

nhis = total number of bins

Calculation, goes inside main loop of the MD code: ideal place routine "lj"

```
do i=1,npart-1
  do j=i+1,npart
    xr = x(i)-x(j)
    xr = xr-box*nint(xr/box)
    r = sqrt(xr**2)
    if (r.lt.box/2) then
      ig = int(r/delg)
      g(ig) = g(ig) +2
    endif
  enddo
enddo
```

!! In "lj" (in 3D)

loop over all pairs

periodic boundary conditions

only within half the box length

Part to add

contribution for particle i and j

↑
endo

Averaging, goes after main loop of the MD code

```
do i=1,nhis
  r=delg*DFLOAT(i + 0.5)
  vb = ((i+1)**3-i**3)*delg**3
  nid = (4/3)*pi*vb*rho
  g(i) =g(i) / (nconf*npart*nid)
enddo
```

```
endif
return
end
```

determine $g(r)$

distance r

volume between bin $i+1$ and i
number of ideal gas part . in vb
normalize $g(r)$

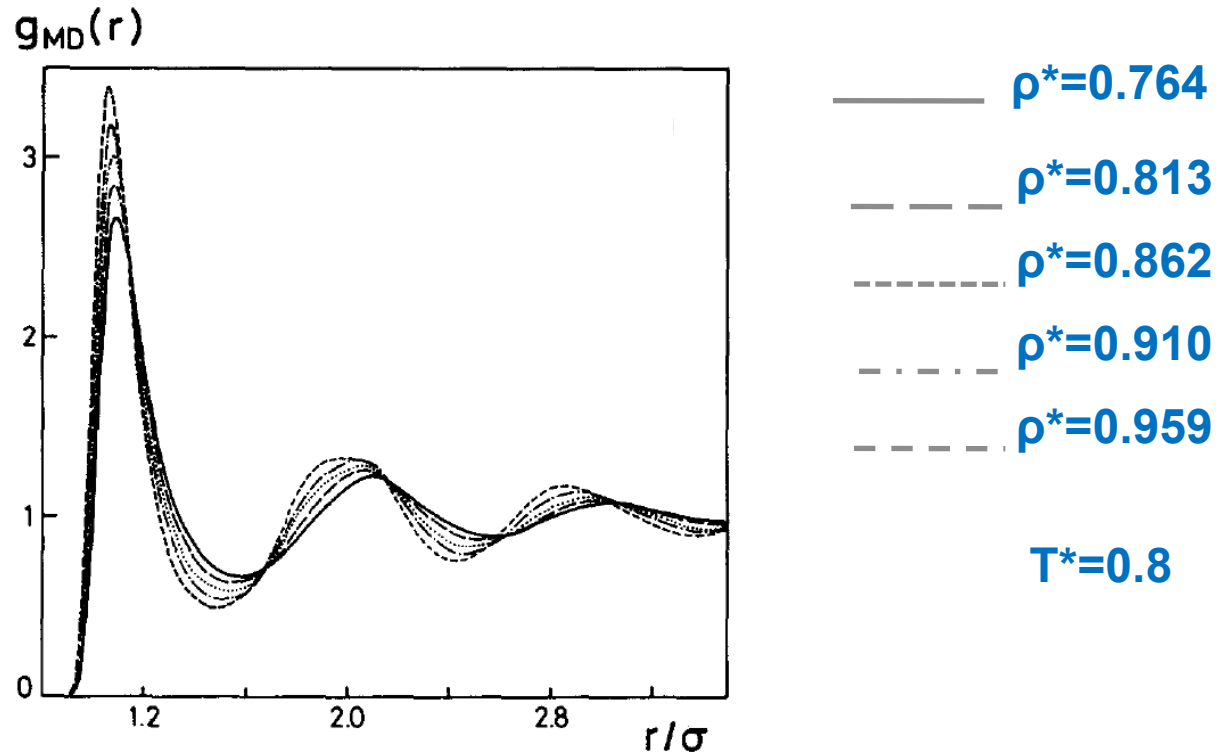
Comments :

- 1. The sampling part is usually combined with the force calculation*
- 2. The factor $\pi=3.14159265358979323846....= 4.d0*DATAN(1.d0)$*

From Frenkel&Smit book



Radial distribution functions for a L-J liquid



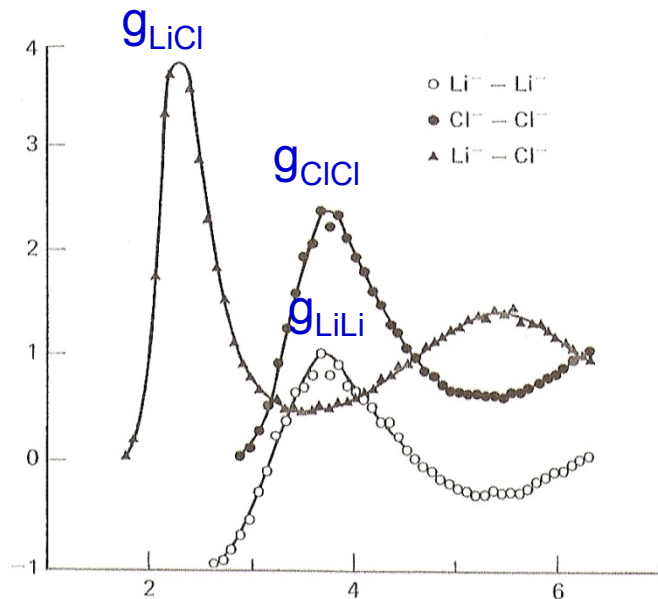
As ρ increases the peaks become more pronounced and move to shorter distances !!

E. Guàrdia et al., J. Chem.Phys. 84, 4569 (1986)

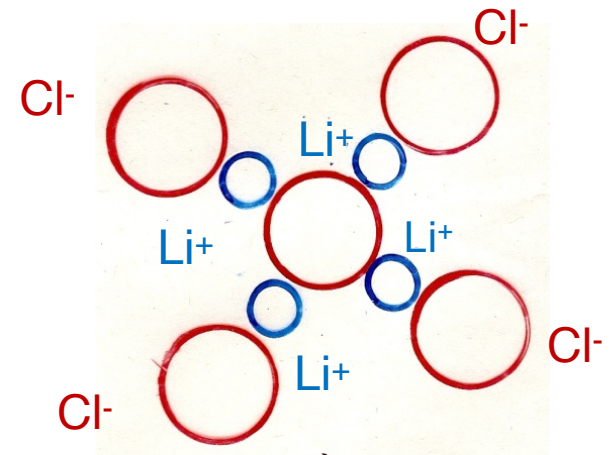
RADIAL DISTRIBUTION FUNCTIONS FOR A MOLTEN SALT

LiCl at $T=883\text{K}$, $V=28.3\text{cm}^3$

Obtained from a Monte Carlo simulation with charged hard sphere potentials



From the positions of maxima and minima:



Coordination numbers:

$$n_{\mu\nu} = x_\nu \int_0^{r_{min}} \rho g_{\mu\nu}(r) 4r^2 dr \quad x_\nu = \frac{N_\nu}{N}$$

H.L.Friedman, "A Course in Statistical Mechanics", pag.81

Why is important the RDF ?

- ✓ It gives information on the **structure**
- ✓ It can be compared with experimental data (neutron diffraction, X-ray scattering)

$$S(k) = 1 + 4\pi\rho_0 \int dr r^2 g(r) \frac{\sin(kr)}{r}$$

($S(k)$ is the *statical structure factor*).

+

- ✓ It can be obtained theoretically by solving integral equations
- ✓ It can be used to compute **thermodynamic properties**

Calculating thermodynamic properties from the RDF

- Energy

$$E = \frac{3}{2}Nk_B T + \langle U_N \rangle$$

$$U_N(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) = \sum_{i=1}^N \sum_{j < i} U(r_{ij})$$

$$\langle U_N \rangle = 2\pi\rho N \int_0^\infty U(r)g(r)r^2 dr$$

- Pressure

$$P = \rho k_B T + \frac{1}{3V} \langle \sum_{i=1}^N \vec{r}_i \cdot \vec{F}_i \rangle$$

$$\vec{F}_i = -\nabla_i U_N(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$$

$$\vec{r}_i \cdot \vec{F}_i = \sum_j r_{ij} \frac{dU(r_{ij})}{dr_{ij}}$$

$$P = \rho k_B T - \frac{2\pi}{3V}\rho^2 \int_0^\infty \frac{dU(r)}{dr} g(r) r^3 dr$$

3. Simulating a system of N particles

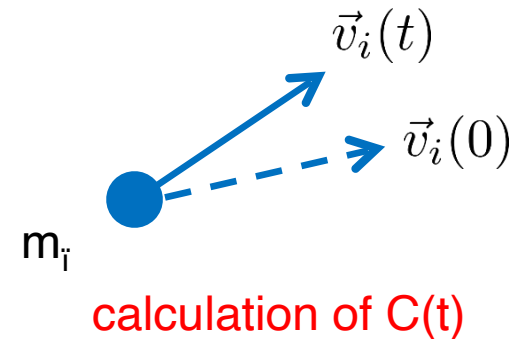
- Dynamic properties: the velocity autocorrelation function VACF

- The velocity autocorrelation function (VACF) describes how velocities vary as a function of time . It can be defined as

$$C(t) = \langle \vec{v}(t) \cdot \vec{v}(0) \rangle$$

$\vec{v}(t)$ Velocity of a particle at time t

$\vec{v}(0)$ Velocity of the same particle at the *initial time* 0



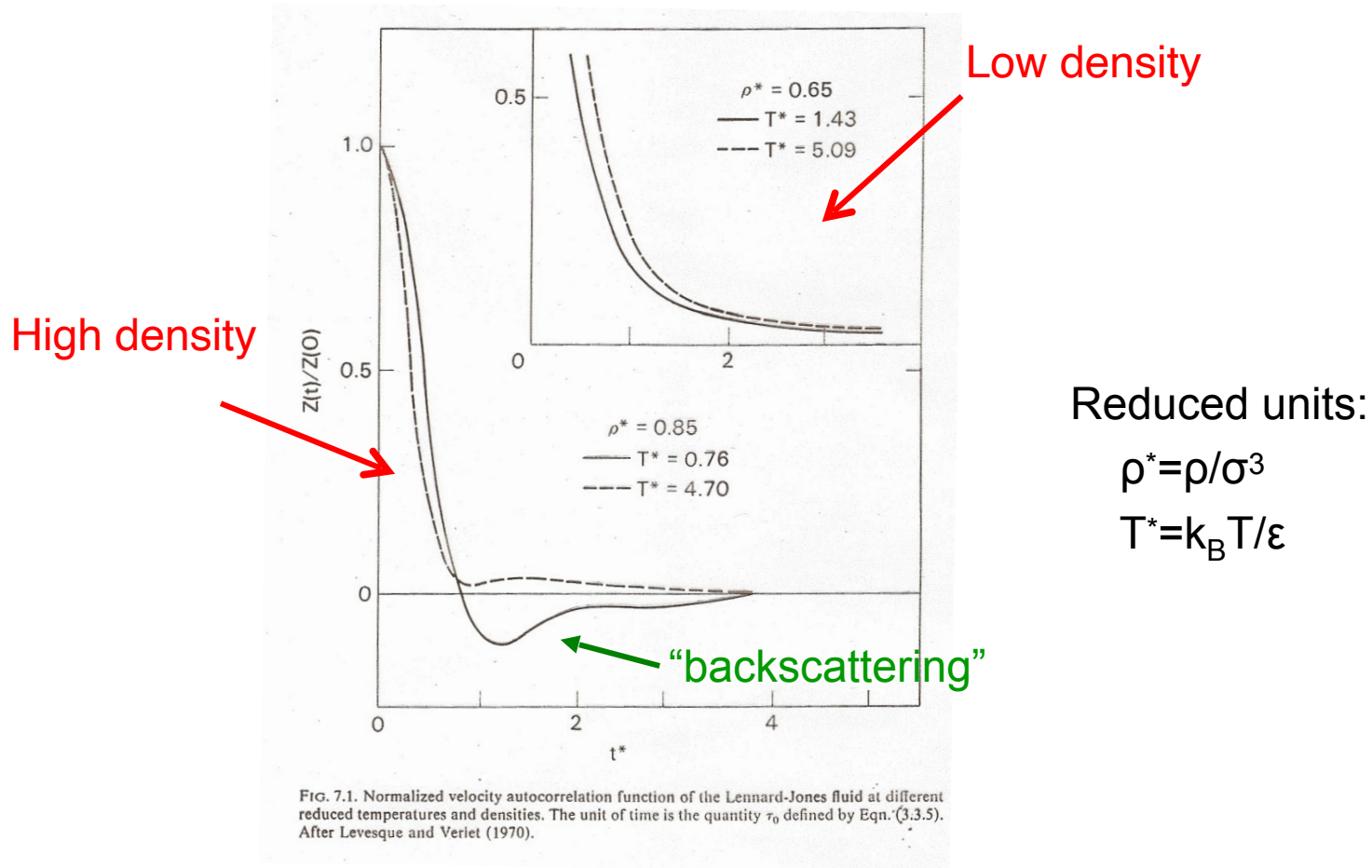
✓ For a monoatomic system, it is

$$C(0) = \langle \vec{v}(0) \cdot \vec{v}(0) \rangle = \langle v^2 \rangle = 3 \frac{k_B T}{m}$$

✓ For an isotropic system, at long times it is

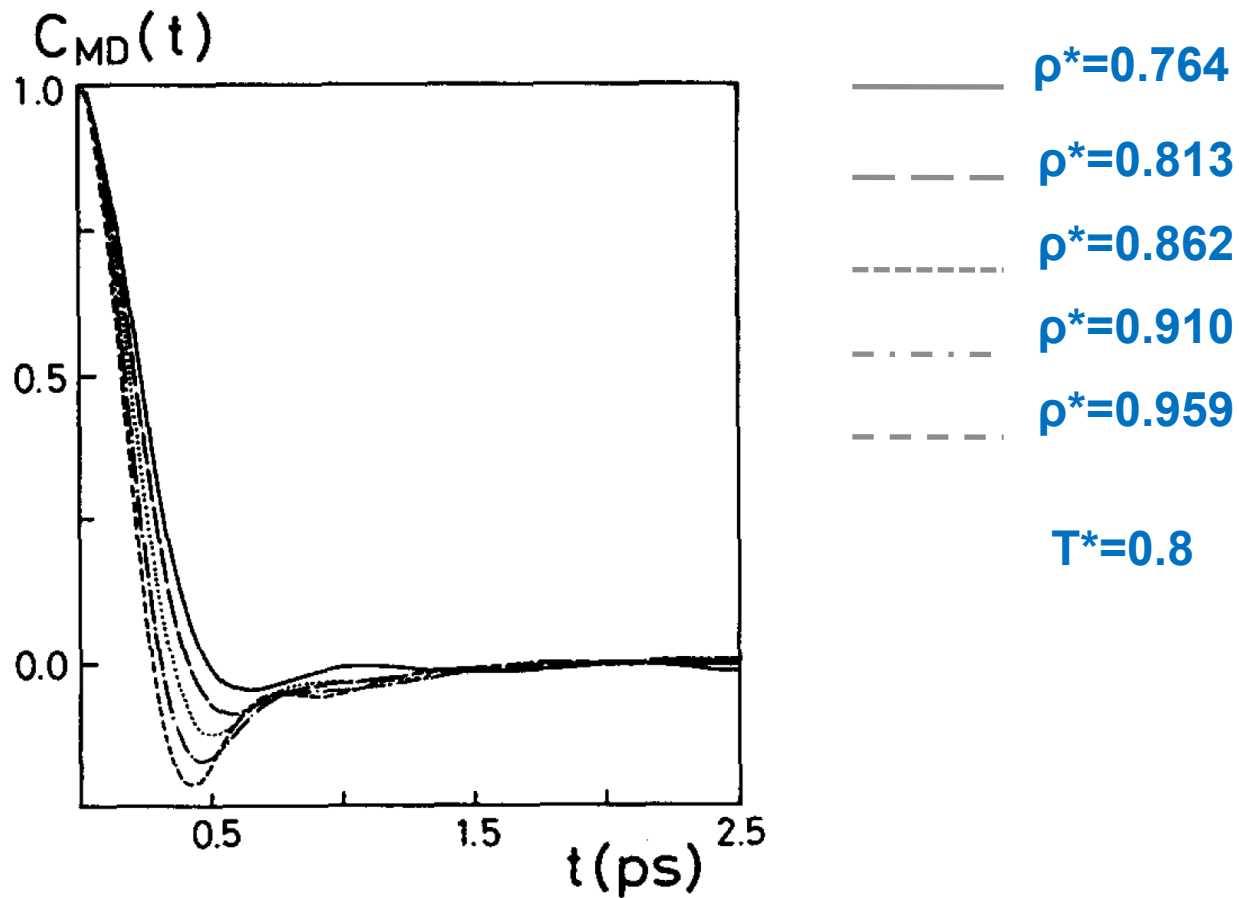
$$\lim_{t \rightarrow \infty} C(t) = \lim_{t \rightarrow \infty} \langle \vec{v}(t) \cdot \vec{v}(0) \rangle = \langle \vec{v} \rangle^2 = 0$$

VELOCITY AUTOCORRELATION FUNCTIONS FOR LIQUID Ar



Hansen & McDonald, "Theory of Simple Liquids", Elsevier 2006

Velocity autocorrelation functions for a dense L-J liquid



As ρ increases the “backscattering” becomes more pronounced !!

E. Guàrdia et al., J. Chem.Phys. 84, 4569 (1986)

Other examples of VACFs

